Wikipedia Web Scraping to Dashboard

Project Title

Country Population Analytics Pipeline

Objective

Build an end-to-end data pipeline that:

- Scrapes population data from Wikipedia
- Cleans and preprocesses the data
- Stores the cleaned data into a database
- Creates an analytical dashboard

Dataset Source

URL:
https://en.wikipedia.org/wiki/List_of_countries_and_dependencies_by_population

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Problem Statement

You are a Data Analyst working for a global research firm. The firm needs an automated system that collects country population data from Wikipedia, cleans it, stores it in a structured database, and visualizes key insights through a dashboard.
Your task is to build this complete pipeline using Python.

Tasks to Perform

Task 1: Web Scraping (Table Extraction)

Using Python, extract the main population table from the provided Wikipedia page.

Requirements

- Use requests and BeautifulSoup (or pandas.read_html)
- Extract the following fields:
  - Country / Territory
  - Population
  - Date
  - Percentage of world population
  - Source (if available)

Expected Output

A raw pandas DataFrame containing the scraped data.

Task 2: Data Cleaning and Preprocessing

Clean the scraped dataset.

Mandatory Cleaning Steps

Students must:

- Remove unwanted symbols (commas, brackets, footnotes)
- Handle missing values
- Convert Population column to numeric
- Standardize country names
- Remove duplicate rows
- Reset index

Deliverable

Cleaned pandas DataFrame ready for analysis.

Task 3: Store Data into Database
Store the cleaned data into a database.
Database Options (choose one)
- SQLite (sqlite3) — Beginner
- MySQL using pymysql — Preferred
Requirements
Students must:
- Create database and table
- Insert cleaned records
- Verify insertion using SELECT query
Table Structure (Suggested)

Column Name	Type
id	INT / INTEGER
country	TEXT / VARCHAR
population	BIGINT
percentage	FLOAT
date	TEXT

Task 4: Exploratory Analysis
Perform basic analysis.
Required Analysis
- Top 10 most populated countries
- Bottom 10 countries
- Average population
- Total world population (from dataset)

Task 5: Dashboard Creation
Create a visual dashboard using:
- matplotlib / seaborn OR
- plotly OR

Required Visualizations
Students must include at least:
1. Bar chart — Top 10 countries by population
2. Pie chart — Population share
3. Horizontal bar — Bottom countries
4. Optional: world population trend (if date usable)